

pod - manage containers.

kubernetes will launch pod not containers.

• Pod - container run as a smallest deployable unit.

- one or more containers - ah kubernetes

manage panha use panha wrapper.

- IP address Irutum for pod.

- Same network and volume for a pod, it may have many containers.
kubernetes, automatically;

1) container run pannum

2) container crash ana restart pannum

3) Multiple containers Irutha manage/scale pannum

4) Traffic - ah containers ku distribute pannum.

5) New version deploy panna help pannum.

* Worker components:

i) pod

ii) container runtime (container d)

iii) kubelet [communication b/w worker & master node]

iv) kube proxy. [used for port forwarding]

7/10/2026

Day-9

Resource Object

Minikube:

• minikube version

• minikube start

Pod commands:

- minikube kubectl -- get nodes |
(can see status, role)

- minikube kubectl -- get pods
↓
used to list

- minikube kubectl -- get pods -A

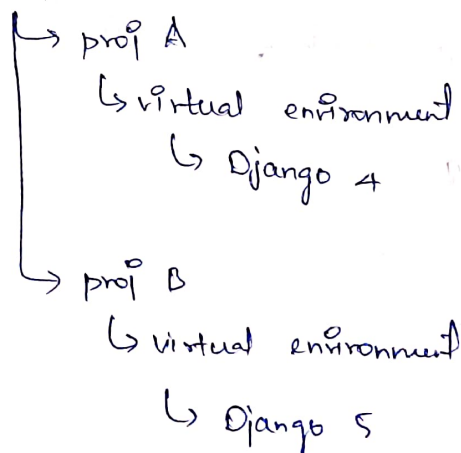
- minikube kubectl -- run apache pod --image=httpd
↙
To create pod & run using httpd docker image
pod name

- minikube kubectl -- describe pod apache pod
↙
To get complete details of pod
(metadata)
pod name

Version Environment

A separate isolated environment created for a project to manage its dependencies independently.

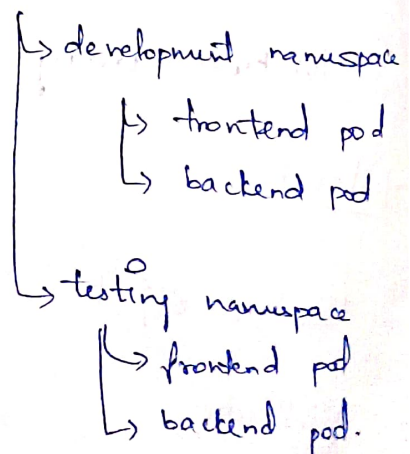
computer



Namespace In kubernetes

cluster kula oru separate working area.

kubernetes cluster



• minikube kubectl --logs apache-pod

↓

pod-kulla running application/container oda logs pakka use pannuvom.

• minikube kubectl --exec -it apache-pod -- /bin/bash
(execute command)

used to enter inside a running pod's container & interact with it using the terminal.

-i ⇒ interactive i/p

t ⇒ terminal allocate panna

/bin/bash ⇒ bash terminal open panna

exec ⇒ pod kulla command execute panna

Commands like apt update, ls, cd /docs, exit.

• minikube kubectl --delete pod apache-pod

{ Declarative method - "Final state epdi irukanum" nu YAML file
Imperative method - "Enna pannanum" nu direct command kudukuvathu. }
[la define pandrathu]

* Resource object file: (declarative YAML file - 4 main sections)

• apiVersion - kubernetes API version

• kind - enna kubernetes resource create panna panna

• metadata - resource ku details kudukum

• spec - resource epdi irukanum nu define pannum.

• minikube kubectl --apply -f simplest.yaml

• minikube kubectl --delete -f simplest.yaml

• minikube kubectl -- apply -f simplepod.yaml

this is used to create or update a kubernetes resource using YAML file.

simplepod.yaml

↓

kubectl apply

↓

kubernetes reads YAML

↓

Creates/updates the Pod

-- ⇒ separates minikube arguments from kubectl command

apply ⇒ create/update according to YAML

-f ⇒ specify a file

simplepod.yaml ⇒ my kubernetes configuration file.

• minikube kubectl -- create deployment ^{name} demodeploy --image=python
image name

• minikube kubectl -- get deploy

• minikube kubectl -- scale deployment demodeploy --replica=5

• minikube kubectl -- get deploy -A

• minikube kubectl -- set image deployment/demodeploy imagename
nginx=nginx:1.14

• minikube kubectl -- rollout status deployment deployname

• " " undo " "

kubernetes - la deployment strategy na, application oda new version -ai old version -lenidhu new version -lu epdi update panna porom nu decide pannadhu.

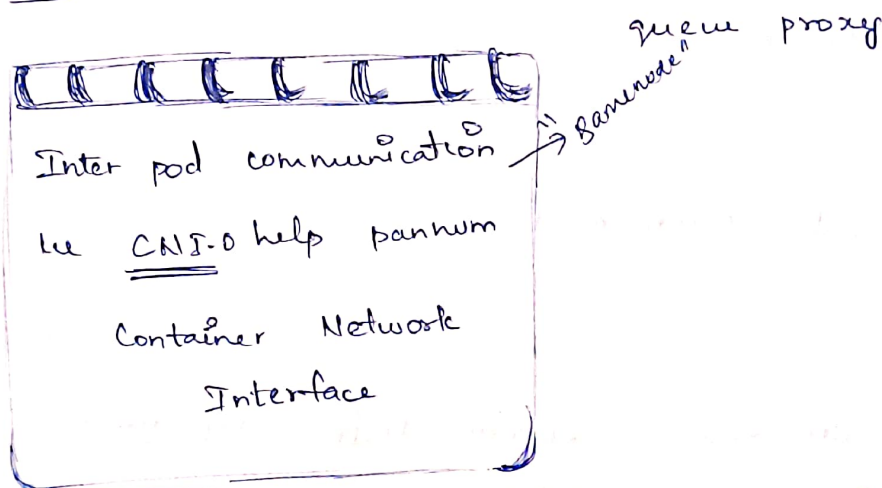
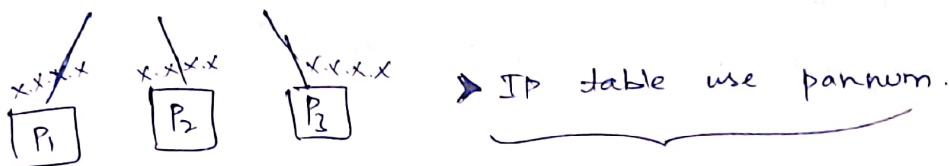
1) Rolling deployment:

Old Pods -ai gradually remove panidu, new Pods, gradually create pannum.

→ Rollout - new version deployment/update process ah rollout nu selluvom.

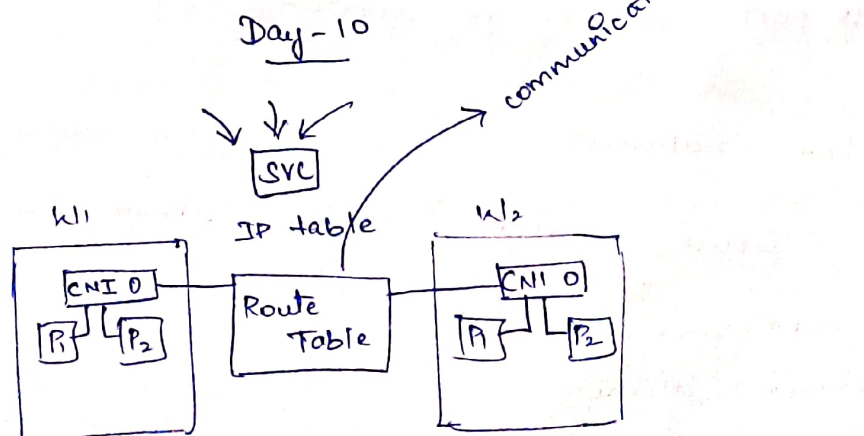
→ Rollback - new version la problem vanthe, previous working version -ku return pannadhu.

• Service resource → static IP kedarkum → service IP.



8/10/2026.

Network



Pod to pod communication - default.

CNI Pluggin

- Flannel
 - Calico
- } N/w policy

* Service resource use podurom for static IP.

* kube proxy generates IP table & it's responsible.
ella pod oda IP irukum.

* Port allocation for kubenetes b/w 20000 - 32767.

* Service to pod communication - IP table.

* Pod to pod communication - CNI o

* CNI O to CNI O (or) 1 pod from 1 node to another
pod in different node - Route table.

4 types of service:

i) Cluster IP:

Internal communication

ii) NodePort:

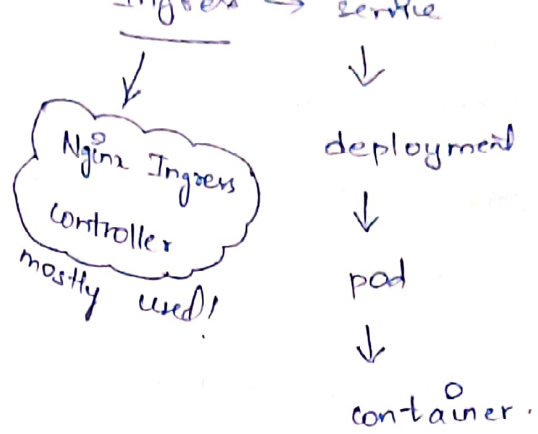
Helps to expose external traffic deployment
in
to port in range 30000-32767

iii) Load balancer:

cloud provider mulima than access pane
mudiyum.

iv) External Name:

Provide records for DNS



- minikube kubectl -- get svc
- minikube kubectl -- get deploy
- minikube kubectl -- get pods
- minikube kubectl -- expose deployment nginxdeploy --type=NodePort --port=80
- minikube kubectl -- scale deployment sornambiga --replicas=5
- minikube service sornambiga
- minikube kubectl -- exec -it sornambiga -- /bin/bash
- minikube kubectl -- delete pod sornambiga

Aws Eluster :

Created a Eluster

sorna-node-group {

- 2) cluster → cluster keela poitu → compute → Add node grp →
- name → permission policy add ~~create~~ → Node IAM role → AmazonEKSRole
- create → No launch template → Next → set compute 2
- scaling config → AMI type - Amazon Linux → Capacity type - On demand
- t3.small → disk size - 20 → Number → next → next → create

3) EC2 → launch → name (master - connect) → AMI (Linux) → security
 grp → select existing SG (SSH, HTTP, HTTPS) = enable na existing
 kudu →

4) master connect → connect → sudo -i → aws configure
→ kops kubectl binary download in browser → Install
other dependencies → linux commands copy → ec2 connect
la paste → kubectl version

5) aws eks update-kubeconfig --name clustername

- kubectl run demopod --image=httpd
- kubectl get pods
- kubectl create deployment deploymentname --image=nginx
demodeploy
- kubectl scale deployment demodeploy --replicas=5
- kubectl get deploy
- kubectl get nodes
- kubectl get pods -o wide ⇒ pod lam enga sen agathu
nu pakalam.
- kubectl expose deployment demodeploy --type=NodePort --port:
- kubectl get svc
- http://IP Public : port no
- kubectl get replica ⇒ command podanum.
- exec command:
 - ls
 - cd
 - ~~cat~~ /usr/share/nginx/html
 - ls
 - touch index.html
 - echo "pod-1" > index.html
 - cat index.html
 - exit